

Public Economics

Lecture 3 · Empirical Tools of Public Finance

Shushanik Margaryan

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WU Vienna University of Economics and Business

Today's Questions

- Theory (Lecture 2) tells us the *direction* government intervention should push behavior. But by how much, in practice?
- How do we know whether a policy *causes* an outcome, rather than merely being *correlated* with it?
- What are “quasi-experiments,” and how can they get us close to the gold standard of a randomized trial when we cannot run one?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 3 — “Empirical Tools of Public Finance”

3.1 Correlation Is Not Causation

3.2 Measuring Causation with Data We'd Like to Have: Randomized Trials

3.3 Estimating Causation with the Data We Actually Get

Structural Modeling

3.4 Conclusion

3.1 Correlation Is Not Causation

The Identification Problem

In the early nineteenth century, a cholera epidemic swept through Russia.

- The government sent doctors to the worst-affected areas.
- Peasants in one province observed a very high **correlation** between the number of doctors in an area and the incidence of cholera there.
- Relying on this fact, they banded together and murdered their doctors.

What went wrong

They correctly saw that doctors and illness moved together — but the *causal* arrow ran the other way: more illness brought more doctors, not the reverse.

Source: Gruber, *Public Finance and Public Policy*, Ch. 3.1.

The Same Mistake Keeps Happening

- September 2014: aid workers trying to help communities in Guinea fight an Ebola outbreak were attacked, and eight were killed. A local leader: *“Wherever those people have passed, the communities have been hit by illness.”*
- 1988: a Harvard dean found that freshmen who had taken SAT-preparation courses scored, on average, *63 points lower* than those who hadn't — and concluded the courses were unhelpful.
- In fact: in 1998 prep courses were rare, so mostly the *weakest* test-takers — who needed the most help — took them. The courses didn't cause worse scores; students likely to score worse were the ones taking the courses.

Statistics calls this the *identification problem*: given that two variables move together, how do you identify what's causing what?

Source: Gruber, *Public Finance and Public Policy*, Ch. 3.1.

Three (and Only Three) Explanations for Any Correlation

For any correlation between two variables A and B :

1. A causes B .
2. B causes A .
3. Some third factor causes both.

Why this matters for policy

Policy makers want to use empirical results to *predict* how an intervention will change behavior. Correlation alone provides no predictive power — prediction requires knowing the *causal* link.

3.2 Measuring Causation with Data

We'd Like to Have: Randomized Trials

The Gold Standard: Randomized Trials

Randomized trial

A group of volunteers is randomly divided into a **treatment group**, which receives the intervention being studied, and a **control group**, which does not — effectively, assignment by the flip of a coin.

- Because assignment doesn't depend on anything about the subjects, the two groups are — on average — identical in *every* respect except the treatment.
- Any remaining difference in outcomes can then be attributed to the treatment: this is what makes randomization the gold standard for estimating causal effects.

The Problem of Bias

Bias

Any source of difference between treatment and control groups that is correlated with the treatment but is *not due to* the treatment.

- Cholera doctors, SAT prep, and breastfeeding studies all suffered from bias: treatment and control groups differed for reasons *other than* the treatment itself.
- By definition, a properly randomized trial has *no* such bias — the only systematic difference between the groups is the treatment.

This is why randomized trials are the benchmark against which every other empirical method is judged.

A Real Randomized Trial: Estrogen Replacement Therapy

- Estrogen replacement therapy (ERT) was widely used to treat menopause symptoms. Early *observational* comparisons found ERT users had no higher (perhaps even lower) heart-disease risk.
- Concern: ERT users were also more likely to be under a doctor's care, live healthier lifestyles, and have higher incomes — all independently linked to lower heart-disease risk. This is textbook *bias* (explanation 3: a third factor drives both).

The Women's Health Initiative trial

A randomized trial of 16,000+ women aged 50–79 across 40 US clinical centers was stopped early (after 5.2 of a planned 8.5 years) because the result was already clear: ERT *raised* heart-disease risk — 7 more coronary events, 8 more strokes, 8 more pulmonary embolisms, and 8 more invasive breast cancers per 10,000 women per year, relative to control.

Source: Gruber, *Public Finance and Public Policy*, Ch. 3.2, citing the Women's Health Initiative trial.

Randomized Trials in Public Finance: The Global Rise of RCTs

- Randomized trials are far more common in medicine than in public finance — but the last two decades have seen a surge, led by the **MIT Jameel Poverty Action Lab (J-PAL)**, founded 2003 by Abhijit Banerjee, Esther Duflo, and Sendhil Mullainathan.
- J-PAL's network has run 1,000+ randomized evaluations of social programs across 91 countries, in partnership with governments, NGOs, and firms.

Example: Indonesia's Raskin subsidized-rice program

Only a third of eligible households received their full rice entitlement, partly because households couldn't prove their eligibility. Researchers randomly gave 378 villages an ID card stating a household's entitlement, versus 194 control villages without one. Card villages saw a sizeable rise in rice actually received — without more going to ineligible households. Indonesia scaled the card nationwide in 2013.

Source: Banerjee, Hanna & Olken, cited in Gruber, *Public Finance and Public Policy*, Ch. 3.2.

A European Randomized Trial: Job-Search Counseling in France

- Public-finance RCTs aren't only a developing-economy story. Behaghel, Crépon & Gurgand (2014) ran a large controlled experiment across roughly 235 French local labor markets ($\approx 44,000$ job seekers).
- Job seekers were randomly assigned to intensive counseling from the **public** employment service, from **private** providers, or to a control group receiving standard services.

Finding

Intensive counseling raised job-exit rates by roughly **15–35%** relative to control — and public-service counseling was about twice as effective as private-provider counseling in the first six months.

A companion study (Crépon, Duflo, Gurgand, Rathelot & Zamora, 2013) used the same randomization to show job-placement help for some can partly *displace* the job chances of others — a caution about reading any single RCT's effect as a free lunch.

Source: Behaghel, Crépon & Gurgand, *AEJ: Applied Economics* 6(4), 2014; Crépon et al., *QJE*, 2013.

Why We Need to Go Beyond Randomized Trials

- **Cost and time:** large trials are expensive and slow to plan and run.
- **Ethics:** a 2001 Parkinson's-disease trial drilled holes in the skulls of all 18 subjects (to keep the control group comparable) but implanted fetal pig cells in only 10 — widely criticized.
- **External validity:** results are only valid for people who *volunteer* to be treatments or controls, who may differ systematically from the population at large.
- **Attrition:** if people leave the trial non-randomly (e.g., those with negative side effects drop out), the remaining sample is no longer representative.

Source: Gruber, *Public Finance and Public Policy*, Ch. 3.2.

3.3 Estimating Causation with the Data We Actually Get

Observational data

Data generated by individual behavior observed in the real world, not in the context of a deliberately designed experiment.

Most of the time, this is all we have. The rest of this lecture covers three approaches researchers use to get *as close as possible* to the gold standard using observational data:

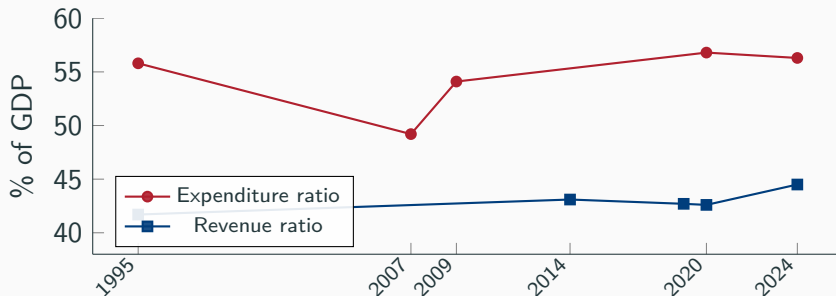
1. Time series analysis
2. Cross-sectional regression analysis
3. Quasi-experiments (incl. difference-in-differences)

Time series analysis

Documenting the correlation between two variables of interest *over time*.

- Simplest possible empirical approach: plot both series across years and see whether they move together.
- **Problem:** when both series follow a slow-moving trend, almost *anything* else that trends over the same period will look correlated with it — the correlation may reflect a shared trend, not causation.

A Time Series You've Already Seen



Discussion (recall Lecture 1)

Both ratios move together around 2009 and 2020. Does this chart tell us the financial crisis or the pandemic *caused* the rise in the expenditure ratio? What is doing the identifying work — the slow trend, or the sharp break?

Source: Fiskalrat, Jahresbericht 2024; Statistik Austria (as in Lecture 1).

When Is Time Series Analysis Useful?

- Slow, gradual trends are *not* convincing — too many other things trend at the same time.
- **Sharp, temporary breaks** are more convincing: if X reverses direction abruptly and Y reverses *at the same moment*, it's harder to find a third factor that just happens to reverse on the identical date.

A sharp break: Ireland's 2002 plastic-bag levy

In March 2002, Ireland introduced a €0.15 levy per plastic shopping bag — one precisely dated policy change. Convery, McDonnell & Ferreira (2007) document a reduction in plastic-bag use “in the order of 90%” at retail outlets; a widely-cited companion figure (Irish government) puts per-capita bag consumption at 328/year before the levy versus 21/year after ($\approx 95\%$).

The drop happens *exactly* at the levy's start date, not gradually — hard to explain with a coincidental third factor. That is what makes a sharp break more convincing than a slow trend.

Source: Convery, McDonnell & Ferreira, “The Most Popular Tax in Europe? Lessons from the Irish Plastic Bags Levy,” *Environmental and Resource Economics* 38(1), 2007.

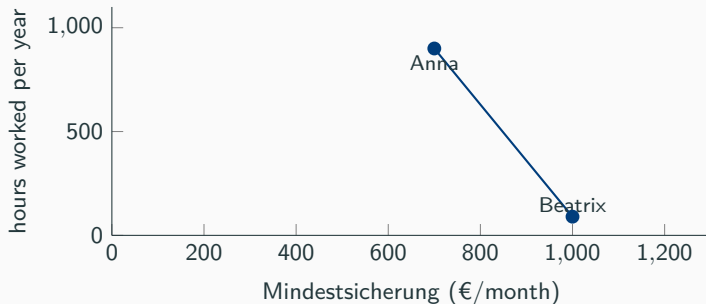
Cross-sectional regression analysis

A statistical method for assessing the relationship between two (or more) variables *while holding other factors constant*, comparing many individuals at *one point in time*.

- In its simplest form (*bivariate regression*), it formalizes correlation: it finds the line that *best fits* a scatter of data points, and reports its slope.
- The **dependent variable** is what you want to explain; the **independent variable(s)** are what you think explain it.

Illustrative Example (Numbers Made Up for Teaching)

Suppose two *Mindestsicherung* (social-assistance) recipients, Anna and Beatrix, have different preferences for leisure versus work.



Anna, with a low preference for leisure, works more *and* — because benefits phase out with earnings — ends up with lower Mindestsicherung. A bivariate regression of hours on benefits would show a strong *negative* slope. **But is this slope causal?**

Problems with Cross-Sectional Regression: Bias Again

- In the Anna/Beatrix example, we know *by construction* that the true story runs the other way: a high preference for leisure causes both *lower* hours worked *and higher* Mindestsicherung (since benefits rise as earned income falls).
- The negative correlation is **not causal** — it reflects a difference in preferences that happens to be correlated with benefit levels.

With real data we don't observe "preference for leisure" directly — but the same bias risk remains: people with a low taste for work may simply be both lower earners *and* higher benefit recipients, with no causal link running from benefits to hours at all.

Control Variables

Control variables

Variables included in a regression to account for differences between comparison groups that could otherwise bias the estimate.

- If we had a perfect measure of “preference for leisure,” we could compare only people with the *same* preference — removing that source of bias entirely.

The catch

The variables we really want (intrinsic preferences, motivation, ability) are usually impossible to measure. We approximate them with proxies — age, education, past employment — which only partially capture the underlying difference. Control variables reduce bias; they rarely eliminate it.

Source: Gruber, *Public Finance and Public Policy*, Ch. 3.3.

Quasi-experiment

A situation that arises naturally when a change in the economic environment (typically, a policy change) creates treatment and control groups that are *nearly identical* except for that change — letting an outside event do the randomizing for us.

Austria's federal structure — nine *Länder*, each able to set some of its own policy parameters (Lecture 1) — is a natural source of quasi-experiments: a reform in one Land, with neighbors unaffected, creates a ready-made treatment/control comparison.

Illustrative Example (Numbers Made Up for Teaching)

Suppose two neighboring Bundesländer, *Vorarlberg* and *Tirol*, both have single-parent Mindestsicherung recipients. In 2023, Vorarlberg cuts its monthly benefit; Tirol's stays unchanged.

	2022	2024	Difference
Vorarlberg (benefit)	€1,000	€800	−€200
Vorarlberg (hours/year)	1,000	1,200	+200
Tirol (benefit)	€1,000	€1,000	€0
Tirol (hours/year)	1,050	1,100	+50

Vorarlberg's raw before/after change (+200 hours) mixes the *benefit cut* with anything else moving nationally (e.g., an economic upswing) — exactly the time series problem. Tirol didn't experience the cut, but *did* experience the same national trend.

Difference-in-difference estimator

The difference between the change in outcomes for the treatment group and the change in outcomes for the control group.

$$\begin{array}{r} \text{hours}(\text{Vorarlberg}, 24) - \text{hours}(\text{Vorarlberg}, 22) = \text{treatment effect} + \text{national-trend bias} \\ \text{hours}(\text{Tirol}, 24) - \text{hours}(\text{Tirol}, 22) = \text{national-trend bias} \\ \hline \text{difference} = \text{treatment effect} \end{array}$$

In the made-up numbers above: $(+200) - (+50) = \mathbf{150}$ additional hours, *net of* the national trend — our diff-in-diff estimate of the causal effect of the benefit cut.

Problems with Quasi-Experimental Analysis

- Diff-in-diff assumes the **parallel-trends** assumption: absent the reform, treatment and control would have moved *the same way*.
- If the national trend hit Vorarlberg and Tirol *differently* (e.g., their labor markets are exposed to different industries), the “bias” terms are not equal — and simple subtraction no longer isolates the true treatment effect.
- Quasi-experiments can never be *certain* they’ve purged all bias, the way a true randomized trial can. Researchers argue their case both intuitively (the groups look comparable) and statistically (checking the result against alternative control groups).

Source: Gruber, *Public Finance and Public Policy*, Ch. 3.3.

Structural Modeling

Reduced-Form vs. Structural Estimates

Reduced-form estimates

Measures of the total impact of one variable on another, without decomposing *why* — what randomized trials and quasi-experiments naturally deliver.

Structural estimates

Estimates of the deeper features (e.g., the utility-function parameters from Lecture 2) that drive the response — letting us decompose an effect into, say, income vs. substitution effects.

Structural models can tell us *why*; but they require assuming a specific form for preferences, and are wrong if that assumption is wrong. This course, like most of the empirical literature, mostly relies on reduced-form evidence. Source: Gruber, *Public Finance and Public Policy*, Ch. 3.3.

3.4 Conclusion

Highlights

- Correlation is not causation: any correlation between A and B could mean $A \rightarrow B$, $B \rightarrow A$, or a third factor driving both.
- Randomized trials remove bias by construction and are the gold standard — but are costly, sometimes unethical, and imperfectly generalizable.
- Time series analysis is convincing only around sharp, dated breaks — not slow trends.
- Cross-sectional regression can reduce, but rarely eliminates, bias — control variables are only as good as the proxies available.
- Quasi-experiments exploit real-world policy changes to approximate randomization; difference-in-differences is the workhorse tool, resting on a parallel-trends assumption.

Discussion Questions for Next Time

1. A researcher finds that Austrian Gemeinden with newly built schools have higher average test scores. Does this prove new schools raise test scores? What else could explain it?
2. Why might a diff-in-difference comparison of two Austrian Länder be more convincing than comparing a single Land's outcomes before and after a reform?
3. Under what conditions would you trust an observational (non-randomized) estimate of a policy's effect nearly as much as a randomized trial?
4. Is it more ethical, or less, to run a randomized trial of a welfare-benefit cut on real recipients than to infer its effect from indirect (observational) evidence alone?
5. Structural models require assuming a specific utility function; reduced-form models don't. What do we gain, and what do we risk, by making that assumption?

Sources

- Textbook: Gruber, J. *Public Finance and Public Policy*, Ch. 3, Worth Publishers — cholera/Ebola/SAT/breastfeeding examples (§3.1); Women's Health Initiative ERT trial and J-PAL/Indonesia Raskin study (§3.2, citing Banerjee, Hanna & Olken); the general logic of Fig. 3-3/3-4 and Table 3-1's illustrative examples (§3.3).
- Behaghel, L., Crépon, B. & Gurgand, M., "Private and Public Provision of Counseling to Job Seekers: Evidence from a Large Controlled Experiment," *American Economic Journal: Applied Economics* 6(4), 2014; Crépon, B., Duflo, E., Gurgand, M., Rathelot, R. & Zamora, P., "Do Labor Market Policies have Displacement Effects?," *Quarterly Journal of Economics*, 2013.
- Convery, F., McDonnell, S. & Ferreira, S., "The Most Popular Tax in Europe? Lessons from the Irish Plastic Bags Levy," *Environmental and Resource Economics* 38(1), 2007.
- Fiskalrat, Jahresbericht 2024; Statistik Austria — Austria expenditure/revenue ratio chart (as in Lecture 1).
- The Anna/Beatrix and Vorarlberg/Tirol examples use **invented, illustrative numbers** for teaching only, following the textbook's own convention (its Fig. 3-3 and Table 3-1 are explicitly "fabricated") — not real Austrian data. A 2016 cross-Länder Mindestsicherung reform did occur, but no published diff-in-diff study of it was found, so none is claimed here.

Thank you!

Questions and discussion